

F3 Definition of a logarithm

Name: _____

Class: _____

Date: _____

Time: **105 minutes**

Marks: **88 marks**

Comments:

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Q1.

On 1 January 1900, a sculpture was valued at £80.

When the sculpture was sold on 1 January 1956, its value was £5000.

The value, £ V , of the sculpture is modelled by the formula $V = A k^t$, where t is the time in years since 1 January 1900 and A and k are constants.

- (a) Write down the value of A . (1)
 - (b) Show that $k \approx 1.07664$. (3)
 - (c) Use this model to:
 - (i) show that the value of the sculpture on 1 January 2006 will be greater than £200 000; (2)
 - (ii) find the year in which the value of the sculpture will first exceed £800 000. (3)
- (Total 9 marks)

Q2.

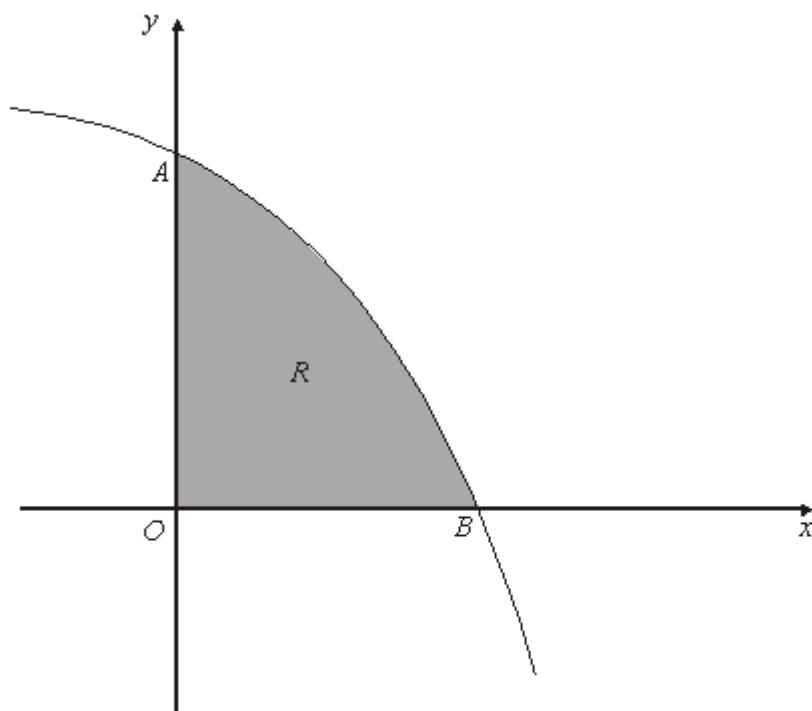
- (a) Given that

$$\log_a x = 2 \log_a 6 - \log_a 3$$
 show that $x = 12$. (3)
 - (b) Given that

$$\log_a y + \log_a 5 = 7$$
 express y in terms of a , giving your answer in a form not involving logarithms. (3)
- (Total 6 marks)

Q3.

The diagram shows a sketch of the curve with equation $y = 27 - 3^x$.



The curve $y = 27 - 3^x$ intersects the y -axis at the point A and the x -axis at the point B .

- (a) (i) Find the y -coordinate of point A . (2)
- (ii) Verify that the x -coordinate of point B is 3. (1)
- (b) The region, R , bounded by the curve $y = 27 - 3^x$ and the coordinate axes is shaded. Use the trapezium rule with four ordinates (three strips) to find an approximate value for the area of R . (4)
- (c) (i) Use logarithms to solve the equation $3^x = 13$, giving your answer to four decimal places. (3)
- (ii) The line $y = k$ intersects the curve $y = 27 - 3^x$ at the point where $3^x = 13$.
Find the value of k . (1)
- (d) (i) Describe the single geometrical transformation by which the curve with equation $y = -3^x$ can be obtained **from** the curve $y = 27 - 3^x$. (2)
- (ii) Sketch the curve $y = -3^x$. (2)

(Total 15 marks)

Q4.

(a) A curve has equation $y = e^{2x} - 10e^x + 12x$.

(i) Find $\frac{dy}{dx}$.

(2)

(ii) Find $\frac{d^2y}{dx^2}$.

(1)

(b) The points P and Q are the stationary points of the curve.

(i) Show that the x -coordinates of P and Q are given by the solutions of the equation

$$e^{2x} - 5e^x + 6 = 0$$

(1)

(ii) By using the substitution $z = e^x$, or otherwise, show that the x -coordinates of P and Q are $\ln 2$ and $\ln 3$.

(3)

(iii) Find the y -coordinates of P and Q , giving each of your answers in the form $m + 12 \ln n$, where m and n are integers.

(3)

(iv) Using the answer to part (a)(ii), determine the nature of each stationary point.

(3)

(Total 13 marks)

Q5.

A function f is defined by $f(x) = 2e^{3x} - 1$ for all real values of x .

(a) Find the range of f .

(2)

(b) Show that $f^{-1}(x) = \frac{1}{3} \ln \left(\frac{x+1}{2} \right)$.

(3)

(c) Find the gradient of the curve $y = f^{-1}(x)$ when $x = 0$.

(4)

(Total 9 marks)

Q6.

A disease is spreading through a colony of rabbits. There are 5000 rabbits in the colony.

At time t hours, x is the number of rabbits infected. The rate of increase of the number of rabbits infected is proportional to the product of the number of rabbits infected and the number not yet infected.

- (a) (i) Formulate a differential equation for $\frac{dx}{dt}$ in terms of the variables x and t and a constant of proportionality k . (2)

- (ii) Initially, 1000 rabbits are infected and the disease is spreading at a rate of 200 rabbits per hour. Find the value of the constant k .

(You are **not** required to solve your differential equation.)

(2)

- (b) The solution of the differential equation in this model is

$$t = 4 \ln \left(\frac{4x}{5000 - x} \right)$$

- (i) Find the time after which 2500 rabbits will be infected, giving your answer in hours to one decimal place. (2)

- (ii) Find, according to this model, the number of rabbits infected after 30 hours. (4)

(Total 10 marks)

Q7.

- (a) Given that

$$\log_a x = 3 \log_a 6 - \log_a 8$$

where a is a positive constant, show that $x = 27$. (3)

- (b) Write down the value of:

(i) $\log_4 1$; (1)

(ii) $\log_4 4$; (1)

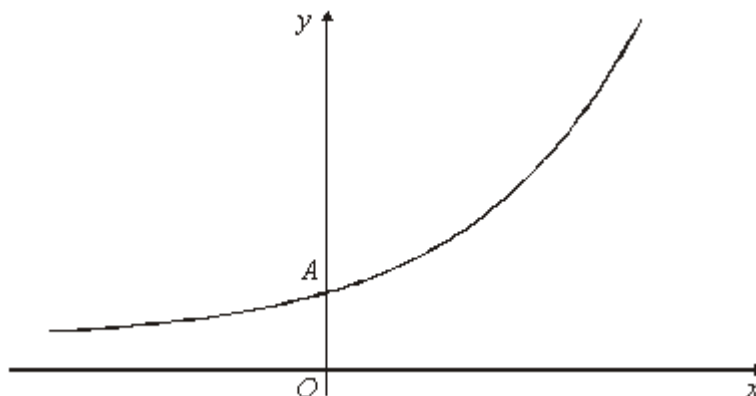
(iii) $\log_4 2$; (1)

(iv) $\log_4 8$. (1)

(Total 7 marks)

Q8.

The diagram shows a sketch of the curve with equation $y = 3^x + 1$.



The curve intersects the y -axis at the point A .

- (a) Write down the y -coordinate of point A . (1)
- (b) (i) Use the trapezium rule with five ordinates (four strips) to find an approximation for $\int_0^1 (3^x + 1) dx$, giving your answer to three significant figures. (4)
- (ii) By considering the graph of $y = 3^x + 1$, explain with the aid of a diagram whether your approximation will be an overestimate or an underestimate of the true value of $\int_0^1 (3^x + 1) dx$. (2)
- (c) The line $y = 5$ intersects the curve $y = 3^x + 1$ at the point P . By solving a suitable equation, find the x -coordinate of the point P . Give your answer to four decimal places. (4)
- (d) The curve $y = 3^x + 1$ is reflected in the y -axis to give the curve with equation $y = f(x)$. Write down an expression for $f(x)$. (1)
- (Total 12 marks)**

Q9.

- (a) Solve the equation $2e^x = 5$, giving your answer as an exact natural logarithm. (2)
- (b) (i) By substituting $y = e^x$, show that the equation $2e^x + 5e^{-x} = 7$ can be written as

$$2y^2 - 7y + 5 = 0$$

(2)

- (ii) Hence solve the equation $2e^x + 5e^{-x} = 7$, giving your answers as exact values of x .

(3)

(Total 7 marks)



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